

Self-test Exercise 7

1. What is the difference between a correlation and a regression analysis?
 - a. Correlation measures the strength and direction of a relationship between two variables, while regression estimates the effect of one variable on another.
 - b. Correlation is used to estimate the mean difference between two groups, while regression estimates the variance of a population.
 - c. Correlation measures the association between two categorical variables, while regression is used for continuous variables.
 - d. Correlation and regression are the same thing.

2. What is the main purpose of instrumental variable analysis?
 - a. To estimate the causal effect of the X on Y correctly, when the sample size goes to infinity
 - b. To adjust for confounding variables so that X and Y relationship can be uncovered.
 - c. To test for interaction effects
 - d. To estimate the prevalence of a disease in a population

3. Consider the following simple regression model: $y_i = \beta_0 + \beta_1 x_i + u_i$, $i = 1, 2, \dots, n$. In order to obtain consistent estimators of β_0 and β_1 when x and u are correlated, a new variable z is introduced into the model which satisfies the following two conditions: $\text{Cov}(z, x) \neq 0$ and $\text{Cov}(z, u) = 0$. The variable z is called a(n) _____ variable.
 - a. dummy
 - b. instrumental
 - c. lagged dependent
 - d. random

4. Consider the following simple regression model: $y = \beta_0 + \beta_1 x + u$. Suppose z is an instrument for x . Which of the following conditions denotes instrument exogeneity?
 - a. $\text{Cov}(z, u) > 0$
 - b. $\text{Cov}(z, x) > 0$
 - c. $\text{Cov}(z, u) = 0$
 - d. $\text{Cov}(z, x) = 0$

5. Consider the following simple regression model $y = \beta_0 + \beta_1 x + u$. Suppose z is an instrument for x . Which of the following conditions denotes instrument relevance?
 - a. $\text{Cov}(z, u) > 0$
 - b. $\text{Cov}(z, u) < 0$
 - c. $\text{Cov}(z, x)$ is not equal to 0
 - d. $\text{Cov}(z, x z) = 0$

6. Consider the following simple regression model $y = \beta_0 + \beta_1 x + u$. Suppose z is an instrument for x . if $\text{Cov}(z, u) = 0$ and $\text{Cov}(z, x) \neq 0$, the value of β_1 in terms of population covariances is _____.
 - a. $\text{Cov}(z, y) / \text{Cov}(z, x)$
 - b. $\text{Cov}(z, u) / \text{Cov}(z, x)$
 - c. $\text{Cov}(z, u)$
 - d. $\text{Cov}(z, x)$

7. The sampling variance for the instrumental variables (IV) estimator is larger than the variance for the ordinary least square estimators (OLS) because _____.

- a. $\rho_{ZX}^2 >$
- b. $\rho_{ZX}^2 < 0$
- c. $\rho_{ZX}^2 = 1$
- d. $\rho_{ZX}^2 < 1$

8. Consider the following simple regression model $y = \beta_0 + \beta_1 x + u$. The variable z is a poor instrument for x if _____.

- a. there is a high correlation between z and x
- b. there is a low correlation between z and x
- c. there is a high correlation between z and u
- d. there is a low correlation between z and u

9. Consider the following simple regression model $y = \beta_0 + \beta_1 x + u$. u and z is an instrument for x . Suppose x and z are both positively correlated with u and $\text{Corr}(z, x) > 0$. Then, the asymptotic bias in the IV estimator is less than that for OLS only if:

- a. $\text{Corr}(z, u) / \text{Corr}(z, x) = \text{Corr}(x, u)$.
- b. $\text{Corr}(z, u) / \text{Corr}(z, x) > \text{Corr}(x, u)$.
- c. $\text{Corr}(z, u) / \text{Corr}(z, x) < \text{Corr}(x, u)$.
- d. $\text{Corr}(z, u) / \text{Corr}(z, x) \neq \text{Corr}(x, u)$.

10. What is the difference between a strong and a weak instrumental variable?

- a. A strong instrumental variable is highly correlated with the endogenous X variable, while a weak instrumental variable has a weaker correlation with the endogenous X variable.
- b. A strong instrumental variable is highly correlated with the dependent variable, while a weak instrumental variable has a weaker correlation with the dependent variable.
- c. A strong instrumental variable is highly correlated with the exogenous X variables that affect the dependent variable, while a weak instrumental variable has a weaker correlation with those variables.
- d. Strong and weak instrumental variables are the same thing.

11. If we focus only on consistency, it is necessarily better to use IV than OLS if the correlation between z and u is smaller than that between x and u .

- a. True
- b. False

12. The two stage least squares estimator is less efficient than the ordinary least squares estimator when the explanatory variables are exogenous.

- a. True
- b. False